

Finding Eigenvalues: Complex Cases

Handling Irrational Roots & 3×3 Matrices

Scenario 1: When you cannot easily factor the equation

Sometimes the characteristic equation $\det(A - \lambda I) = 0$ does not factor into nice integers. In these cases, we must use the **Quadratic Formula**.

Example

Find eigenvalues for $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$.

Step 1: Characteristic Equation

$$\det \begin{bmatrix} 1 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix} = 0$$

Step 2: Expand

$$\begin{aligned}(1 - \lambda)(3 - \lambda) - (2)(1) &= 0 \\ (3 - \lambda - 3\lambda + \lambda^2) - 2 &= 0 \\ \lambda^2 - 4\lambda + 1 &= 0\end{aligned}$$

Step 3: Apply Quadratic Formula For $a\lambda^2 + b\lambda + c = 0$, the roots are $\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Here, $a = 1, b = -4, c = 1$.

$$\begin{aligned}\lambda &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)} \\ \lambda &= \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm \sqrt{12}}{2}\end{aligned}$$

Simplify $\sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3}$:

$$\lambda = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Solution

The eigenvalues are irrational:

$$\lambda_1 = 2 + \sqrt{3}, \quad \lambda_2 = 2 - \sqrt{3}$$

Scenario 2: The 3×3 Matrix

For a 3×3 matrix, the determinant calculation requires **Cofactor Expansion**.

Example

Find eigenvalues for $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$.

Step 1: Set up $A - \lambda I$ Subtract λ from the diagonal elements only.

$$A - \lambda I = \begin{bmatrix} 4 - \lambda & 0 & 1 \\ -2 & 1 - \lambda & 0 \\ -2 & 0 & 1 - \lambda \end{bmatrix}$$

Step 2: Find Determinant (Pick the easiest row/column) *Strategy: Choose the row or column with the most zeros to simplify calculation.* Here, **Column 2** has two zeros. We expand along Column 2.

The sign pattern for a 3×3 is $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$. For Col 2, the signs are $(-, +, -)$.

$$\det(A - \lambda I) = \underbrace{-0(\dots)}_{\text{Row 1}} + \underbrace{(1 - \lambda) \cdot \det \begin{bmatrix} 4 - \lambda & 1 \\ -2 & 1 - \lambda \end{bmatrix}}_{\text{Row 2}} - \underbrace{0(\dots)}_{\text{Row 3}}$$

Step 3: Calculate the 2×2 Determinant inside

$$\begin{aligned} \det \begin{bmatrix} 4 - \lambda & 1 \\ -2 & 1 - \lambda \end{bmatrix} &= (4 - \lambda)(1 - \lambda) - (1)(-2) \\ &= (4 - 4\lambda - \lambda + \lambda^2) + 2 \\ &= \lambda^2 - 5\lambda + 6 \end{aligned}$$

Step 4: Combine and Solve Substitute this back into our expansion from Step 2:

$$(1 - \lambda) \cdot [\lambda^2 - 5\lambda + 6] = 0$$

We now have two parts to solve:

1. Part A: $1 - \lambda = 0 \implies \lambda = 1$

2. Part B: $\lambda^2 - 5\lambda + 6 = 0$

Factor Part B: $(\lambda - 2)(\lambda - 3) = 0 \implies \lambda = 2, \lambda = 3$.

Final Answer

The eigenvalues are:

$$\lambda = 1, \quad \lambda = 2, \quad \lambda = 3$$